Knowledge Graphs of Bank Volatility in the COVID-19 Crisis *

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1 Introduction

A knowledge graph represents entities as vertices and their relationships as edges. Chen and Zhang (2024) applied this concept to assess systemic risk in U.S. banking. They estimated a sparse precision matrix over the leading financial firms, treating non-zero off-diagonal entries as edges. They compared the graph during the 2008 financial crisis to the graph outside of that period. Regarding the liquidity discount index (Chen et al., 2016), they found that firms were more closely interconnected during the crisis, which supports the idea that a common macroeconomic shock displaces idiosyncratic risk. Regarding volatility, they reported that the non-crisis period produced a more complex network than the crisis period.

This project extends their framework to the pandemic episode of 2020. We follow 70 banks listed in the United States at daily frequency from 2017 to 2021, and compare the graph inside the pandemic window with the graph outside it. At a common cut-off $c = 0.035$, we found that none of the banks are isolated during the crisis, and the network holds a dense core. However, this claim applies only to weak links. A large share of strong conditional dependence in this sample occurs in the non-crisis period. At $c = 0.050$, there are 8 edges inside the crisis against 84 outside it, and the surviving structure matches the geographic split of the non-crisis graph. Section 2 describes the data. Section 3 applies the embeddings and Section 4 estimates the knowledge graph. Section 5 concludes.

2 Data

The universe is 70 bank stocks listed in the United States, taken from the ranking by market capitalization published by *Stock Analysis*. Daily adjusted closing prices were collected from May 2017 to December 2021. We define volatility as the trailing standard deviation over 1 year, so the study window runs from June 2018 to December 2021. The crisis window is defined as December 2019 to June 2020, following the acute phase of the market shock identified by Liu et al. (2020), and the non-crisis window is the remainder of the study period. Table 1 states the windows, and Table 2 reports summary statistics.

Period	Start	End	Observations
Full study	June 2018	December 2021	883
Crisis	December 2019	June 2020	124
Non-crisis	June 2018	November 2019	759
	June 2020	December 2021	

Table 1: Definition of the subperiods.

3 Embedding

Write $\sigma_{i,t}$ as the rolling sample volatility, defined as the trailing standard deviation over 1 year (252 trading days). Denote $z_{i,t} = s_i^{-1}(\sigma_{i,t} - \bar{\sigma}_i)$ be the standardized volatility within each

*Code and replication data are available at *GitHub*.

Year	Mean (%)	Median (%)	Volatility (%)
2017	+17.5	+11.9	22.3
2018	-22.0	+0.0	27.3
2019	+18.2	+30.4	29.0
2020	-12.7	-8.5	62.0
2021	+18.5	+14.9	31.2

Table 2: Summary statistics of daily log returns, annualized.

subsample. We first apply three embeddings to map the resulting panel into two dimensions, namely locally linear embedding, spectral embedding and principal component analysis. The first two are eigenmaps that preserve local geometry, and the third is the linear baseline. A k-means partition with $k = 5$ colors the vertices and serves as an independent check on the embedding, as in Chen and Zhang (2024).

Figure 1 shows the three embeddings of all 70 banks over the study window. The Locally Linear Embedding (LLE) visualization reveals several distinct clusters of institutions. Most firms cluster in a dense mass around (0.0, 0.0), with only a few outliers positioned separately. It demonstrates that when most firms exhibit similar volatility dynamics over the entire period, LLE maps them to nearly identical coordinates. In contrast, spectral embedding provides a more informative layout, revealing a curved manifold structure that suggests nonlinear relationships among the banks' volatility patterns. This visualization also displays clearly separated clusters along the arms of the shape. Principal component analysis (PCA) is the most dispersed of the three and spreads the banks across both axes.

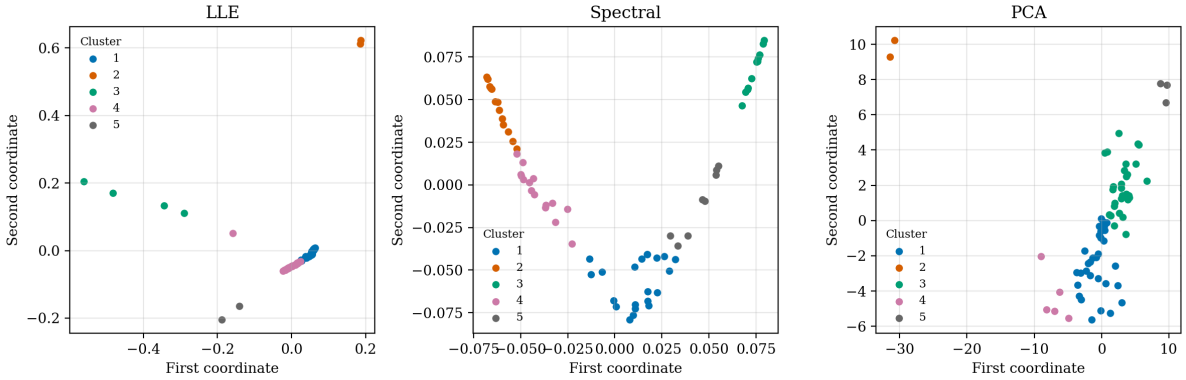


Figure 1: Embeddings of the standardized volatility of all 70 banks over the study window, colored by a k-means partition with $k = 5$.

4 Graphical LASSO

Next, we utilize the graphical LASSO, a precision-matrix LASSO, to identify conditional independence. Let S be the sample correlation matrix of $\{z_{i,t}\}$ over a subsample. Graphical LASSO estimates

$$\hat{\Theta} = \arg\min_{\Theta \succ 0} \{\operatorname{tr}(S\Theta) - \log\det\Theta + \alpha\|\Theta\|_1\}, \tag{1}$$

where the penalty α is chosen by three-fold cross-validation. A non-zero off-diagonal entry means that two banks remain dependent once every other bank in the cross section is conditioned on. Note that the entries of $\hat{\Theta}$ are not comparable across two separate fits, since each carries the

scale of its own diagonal. The edge weight is therefore the partial correlation

$$\hat{\rho}_{ij} = -\frac{\hat{\Theta}_{ij}}{\sqrt{\hat{\Theta}_{ii}\hat{\Theta}_{jj}}} \in [-1, 1], \qquad E(c) = \#\left\{(i,j) : i < j,\ |\hat{\rho}_{ij}| > c\right\}, \tag{2}$$

where c is a cut-off. The 25 largest banks are carried into the spring layout, for visualization, edge width is proportional to $|\hat{\rho}_{ij}|$ and vertex color is the cluster label.

Table 3 reports the selected penalty and partial-correlation statistics. Cross-validation selects $\alpha = 0.600$ inside the crisis and $\alpha = 0.599$ outside it, the two fits are therefore penalized to the same degree. The two subsamples have nearly the same median partial correlation, 0.0397 versus 0.0386. However, the standard deviation outside the crisis is three times as large as during the crisis.

	Crisis	Non-crisis
Selected penalty α	0.600	0.599
Median $\lvert\hat{\rho}_{ij}\rvert$	0.0397	0.0386
Standard deviation of $\lvert\hat{\rho}_{ij}\rvert$	0.0061	0.0199

Table 3: Fitted precision matrices of the 25 largest banks.

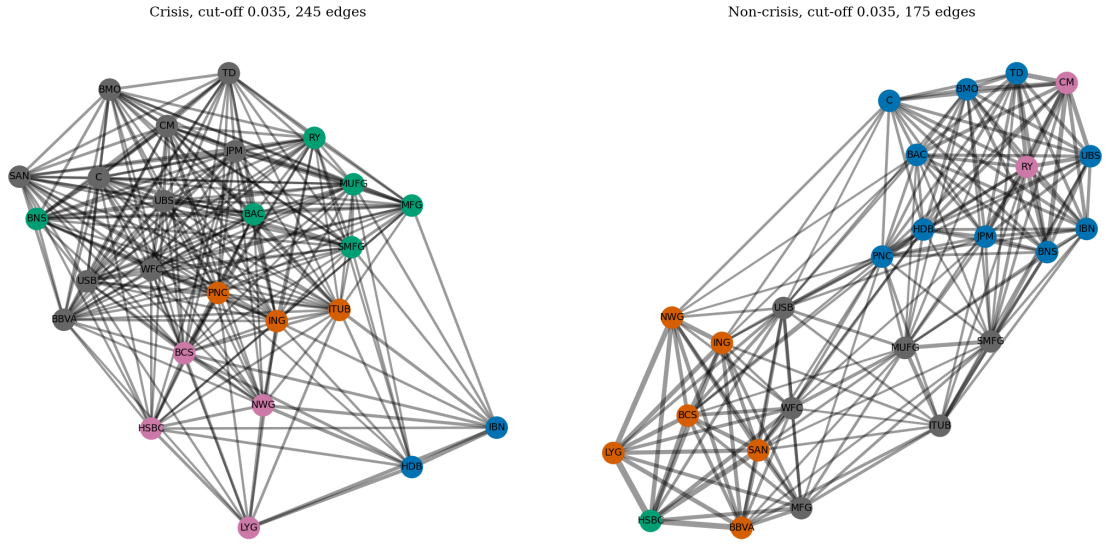


Figure 2: Knowledge graphs of the 25 largest banks at the common cut-off $c = 0.035$.

Figure 2 draws the 25 largest banks at the common cut-off $c = 0.035$. The crisis graph carries 244 of the 300 possible edges, and the non-crisis graph carries 175. No bank is isolated during the crisis. This finding aligns with Chen and Zhang (2024), which indicates that firms tend to become more interconnected during crisis periods when common macroeconomic shocks overshadow idiosyncratic risks in liquidity measures. The network holds a dense core of North American and continental European names and leaves the Indian and British listings, IBN, HDB and LYG, on a tail at the edge. The cluster colors are scattered across that core rather than aligned with it, so the clustering does not corroborate the graph inside the crisis.

The non-crisis graph is visually sparser and more organized. It separates into a North American group of {C, BAC, BMO, TD, CM, RY, UBS, JPM, BNS and PNC} together with the two Indian listings, and a British and continental European group of {NWG, ING, BCS,

LYG, HSBC, SAN and BBVA}, joined through {USB, WFC} and the Japanese and Brazilian names {MUFG, SMFG, MFG and ITUB}. HSBC and LYG are the least connected at nine edges each. This aligns with HSBC's hybrid role, operating at the intersection of Asian and European markets, hence resulting in a volatility profile that does not attach strongly to either group. Here, the cluster colors follow the two groups, so the clustering corroborates the layout, which Chen and Zhang (2024) assign an auxiliary role.

Compared with the crisis graph, several edges that were thick during the crisis have completely disappeared in non-crisis times. This indicates that shock transmission between institutions is less pronounced in periods of stability. Figure 3 carries the explanation. The two edge counts cross at $c = 0.040$, and below the crossing the crisis graph is denser, at $c = 0.020$ it is almost complete at 298 edges against 253. Above the crossing, the ordering reverses and widens. At $c = 0.050$ the crisis graph holds 8 edges and the non-crisis graph 84, and at $c = 0.075$ the figures are 1 and 8. The claim that banks are more connected during the crisis is therefore a claim about weak links only. A high proportion of strong conditional dependence in this sample belongs to the non-crisis period, where the surviving edges resolve into a North American block and a European block joined through the Japanese and Brazilian names. This matches the volatility result of Chen and Zhang (2024), which they describe as counterintuitive relative to their liquidity result.

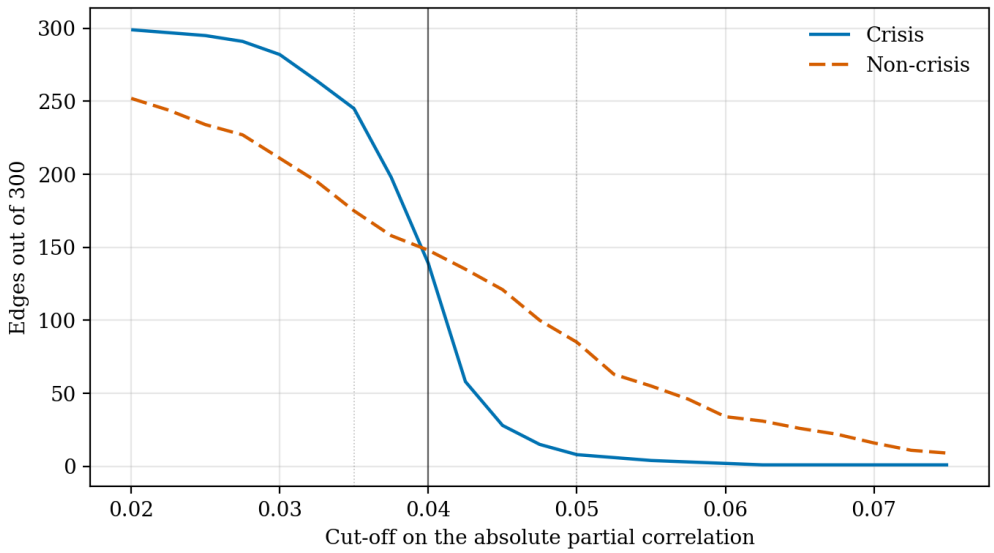


Figure 3: Edges against a common cut-off on the absolute partial correlation. The solid vertical line marks the crossing at $c = 0.040$ and the dotted lines mark $c = 0.035$ and $c = 0.050$.

5 Conclusion

This project transfers the framework of Chen and Zhang (2024) to the pandemic episode. At $c = 0.035$, we found that none of the 25 largest banks are isolated during the crisis, and the network holds a dense core. This aligns with the expectation that firms tend to become more interconnected during crisis periods when common macroeconomic shocks overshadow idiosyncratic risks. Nevertheless, this is a claim about weak links only. A large share of strong conditional dependence in this sample occurs in the non-crisis period. At $c = 0.050$, there are 8 edges inside the crisis against 84 outside it, and the surviving structure matches the geographic split of the non-crisis graph, consistent with the volatility result of Chen and Zhang (2024).

References

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